Mixed benchmarks from earlier units Student code: _____ Calculator: yes

1. The price of one share of a company's stock changed by these amounts over 4 days.

−$0.60, +$1.55, −$1.95, −$0.40

What was the mean (average) daily change in the price?

2. A pair of shoes costs $27.11 and is on sale for 37.5% off.

What is the sale price, to the nearest cent?

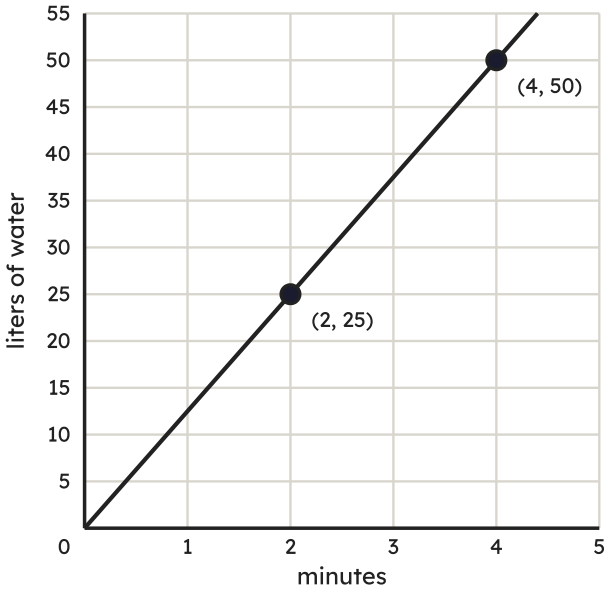
3. What is the exact value of m in the equation shown?

$$-3\frac{19}{25} = \left(\frac{3}{5}\right)m - 4$$

Turn over →

4. The graph shows the proportional relationship between minutes and liters of water.

What is the constant of proportionality, in liters per minute?



5. A right circular cylinder has a radius of 4 units and a height of 6 units.

What is the volume of the cylinder, in cubic units?

A. 48π

B. 24π

C. 192π

D. 96π

Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — BACKSTOP-Q4

Given when no new check is ready by Thursday. Logged as a real check (Backstop = y in CHECKS).

1. MA.7.NSO.2.3 Regular **−$0.35**

Add the 4 changes: the total change is −$1.40.

Divide by the number of days: $-\$1.40 \div 4 = -\$0.35$.

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2. MA.7.AR.3.1 Regular **$16.94**

Discount: $\$27.11 \times 0.375 = \$10.16625 \approx \$10.17$.

Sale price: $\$27.11 - \$10.17 = \$16.94$.

37.5% must be written as 0.375 before multiplying.

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3. MA.7.AR.2.2 Regular $m = \dfrac{2}{5}$

Add 4 to both sides: $-3\dfrac{19}{25} + 4 = \dfrac{6}{25}$, so $\dfrac{6}{25} = \left(\dfrac{3}{5}\right) m$.

Divide both sides by $\dfrac{3}{5}$, which means multiplying by its reciprocal $\dfrac{5}{3}$: $m = \dfrac{6}{25} \times \dfrac{5}{3} = \dfrac{2}{5}$.

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4. MA.7.AR.4.2 Regular **12.5 liters per minute**

Use a labeled point: $(2,\ 25)$.

The constant of proportionality is $y \div x = 25 \div 2 = 12.5$ liters per minute.

It is not the y-value of a labeled point, because no labeled point has $x = 1$.

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5. MA.7.GR.2.3 Regular **D.** 96π

$V = \pi r^2 h = \pi(4)^2(6) = 96\pi$.

The value 48π would be the curved surface area instead.
